

Matrix Inverse

Geometric Algorithms
Lecture 10

Practice Problem

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 2x_1 - x_2 \\ 4x_2 \\ 3x_1 - 3x_2 \end{bmatrix}$$

*Determine if the above transformation is onto,
one-to-one, both, or neither*

YES

NO

Answer

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} -1 \\ 4 \\ -3 \end{bmatrix}$$

$$T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 2x_1 - x_2 \\ 4x_2 \\ 3x_1 - 3x_2 \end{bmatrix}$$

$\mathbb{R}^2$ $\mathbb{R}^3$

1-1 because
col. are L.I.

$$T(\vec{x}) = \begin{bmatrix} 2 & -1 \\ 0 & 4 \\ 3 & -3 \end{bmatrix} \vec{x}$$

3×2 $\mathbb{R}^2$

$$\begin{bmatrix} 2 & -1 \\ 0 & 4 \\ 3 & -3 \end{bmatrix}$$

not co-linear, L.I

not onto because
the matrix
is tall (more
rows
than
cols)

Objectives

- » Define other several important matrix operations
- » Motivate and define matrix inverses

Keywords

Matrix Transpose

Inner Product

Matrix Power

Square Matrix

Matrix Inverse

Invertible Transformation

1-1 Correspondence

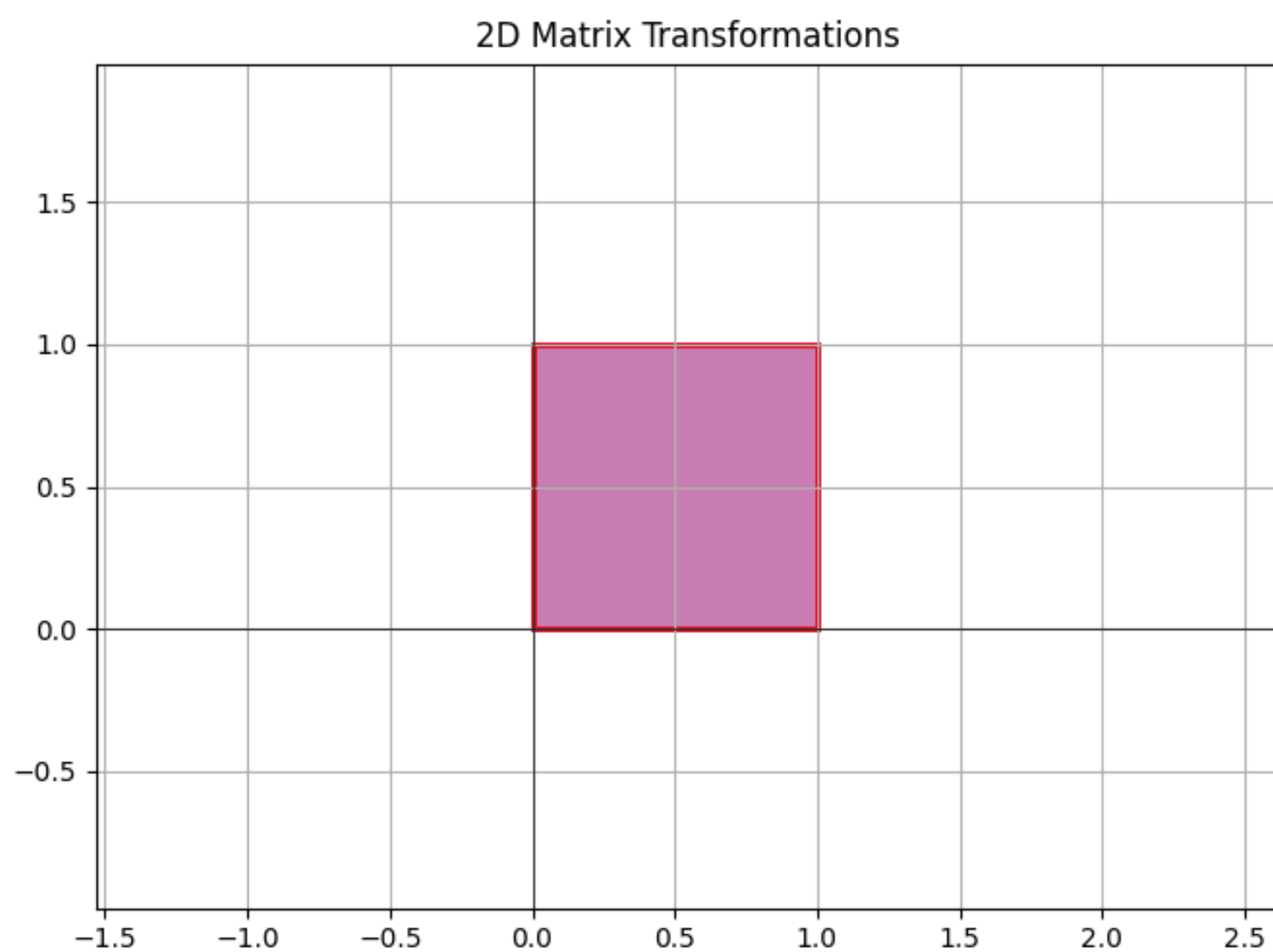
`numpy.linalg.inv`

Determinant

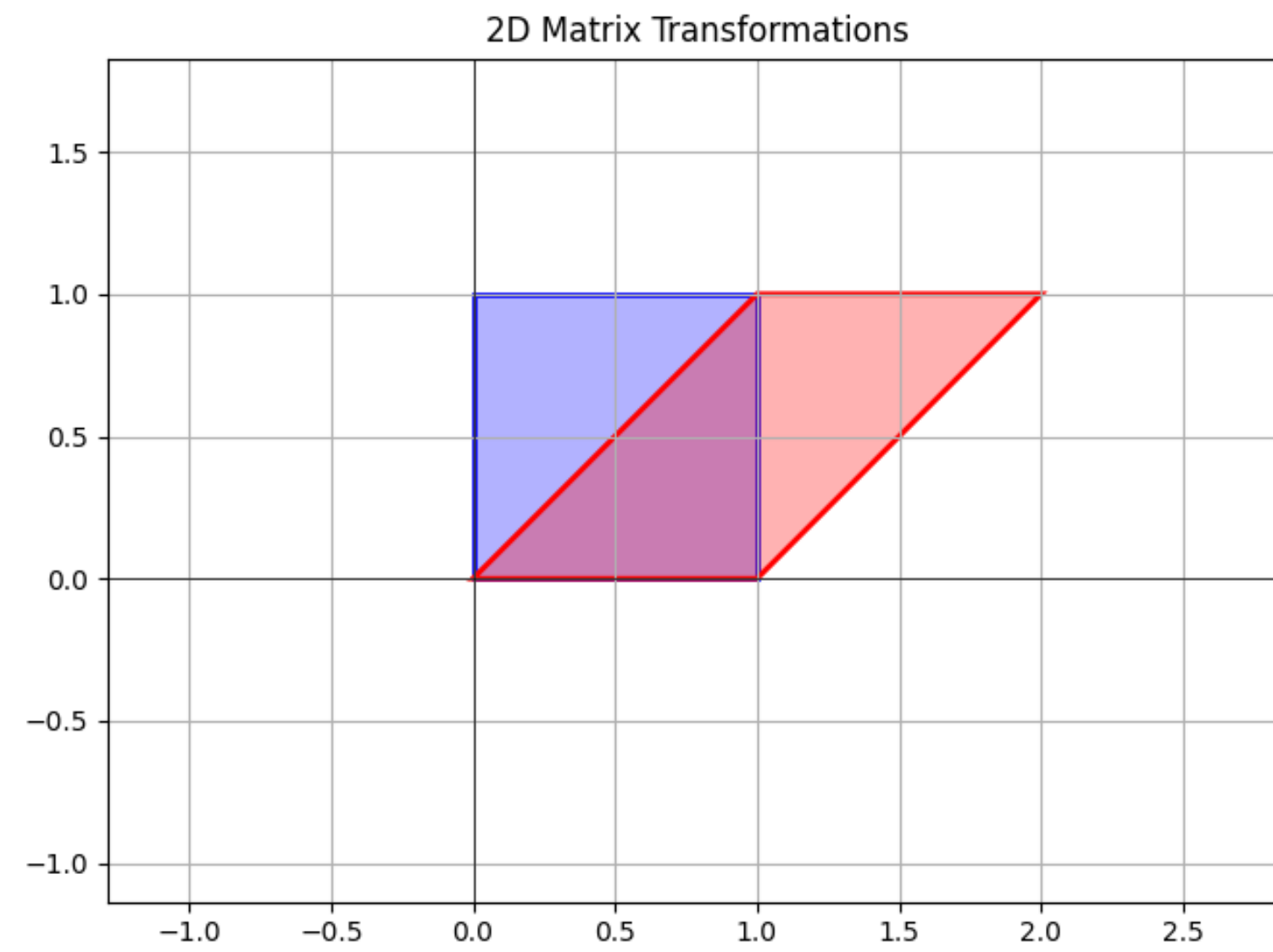
Invertible Matrix Theorem

Recap

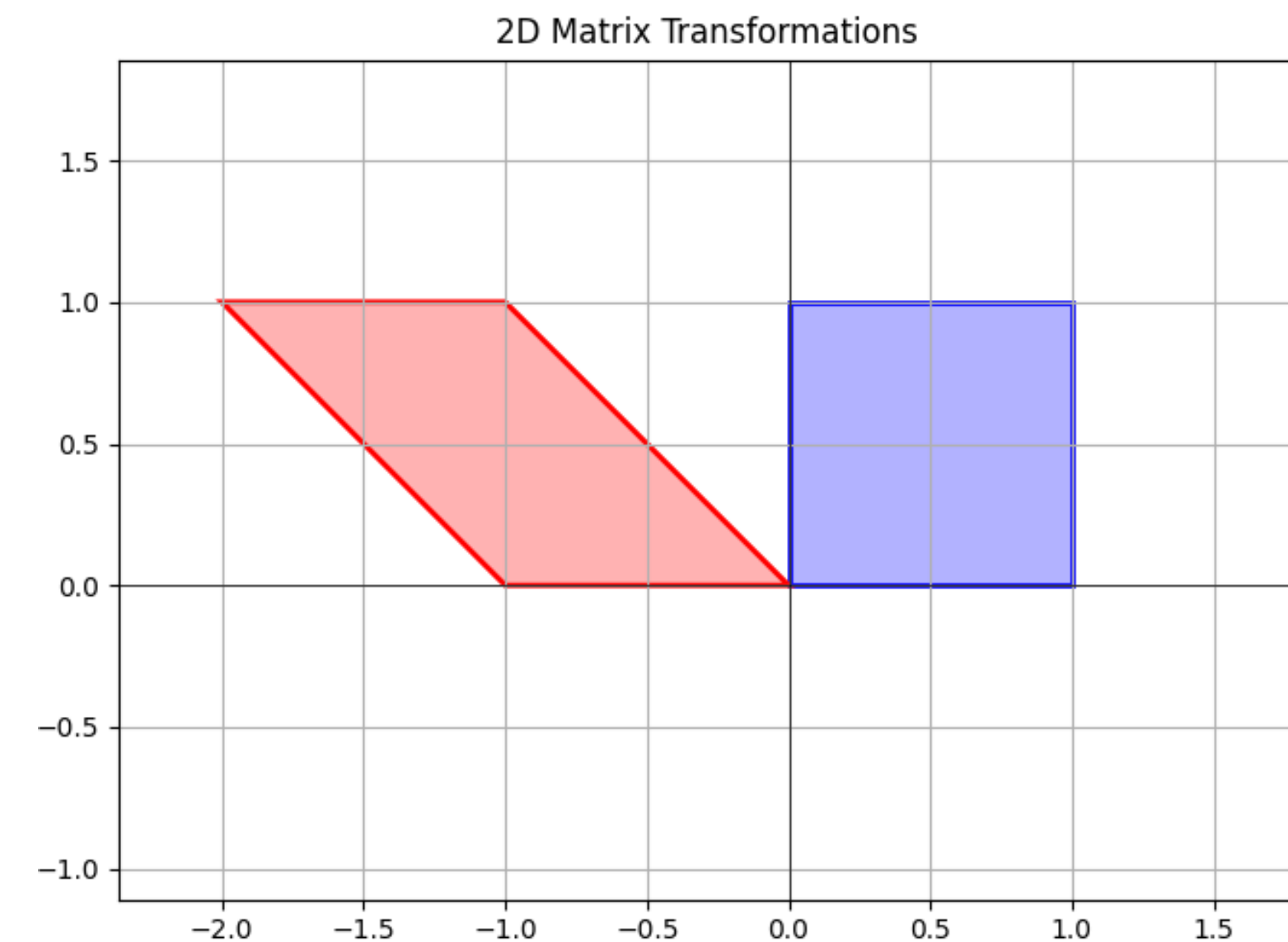
Recap: Shearing and Reflecting (Geometrically)



shear



reflect



Recap: The Key Fact

Thm. The composition of two linear transformations is a linear transformation

This means the composition of two matrix transformations can be implemented by a *single* matrix

Matrix Multiplication

Def. For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ with columns $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$ the product $AB \in \mathbb{R}^{m \times p}$ is given by

$$AB = A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \dots & A\mathbf{b}_p \end{bmatrix}$$

Replace each column of B with A multiplied by that column

Recap: Matrix Multiplication and Composition

Thm. For any matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ and any vector $\mathbf{v} \in \mathbb{R}^p$,

$$A(B\mathbf{v}) = (AB)\mathbf{v}$$

The matrix implementing the composition is the product of the two underlying matrices

Recap: Matrix "Interface"

multiplication

what does AB mean when A and B are matrices?

addition

what does $A + B$ mean when A and B are matrices?

scaling

what does cA mean when A is matrix and c is a real number?

More Matrix Operations

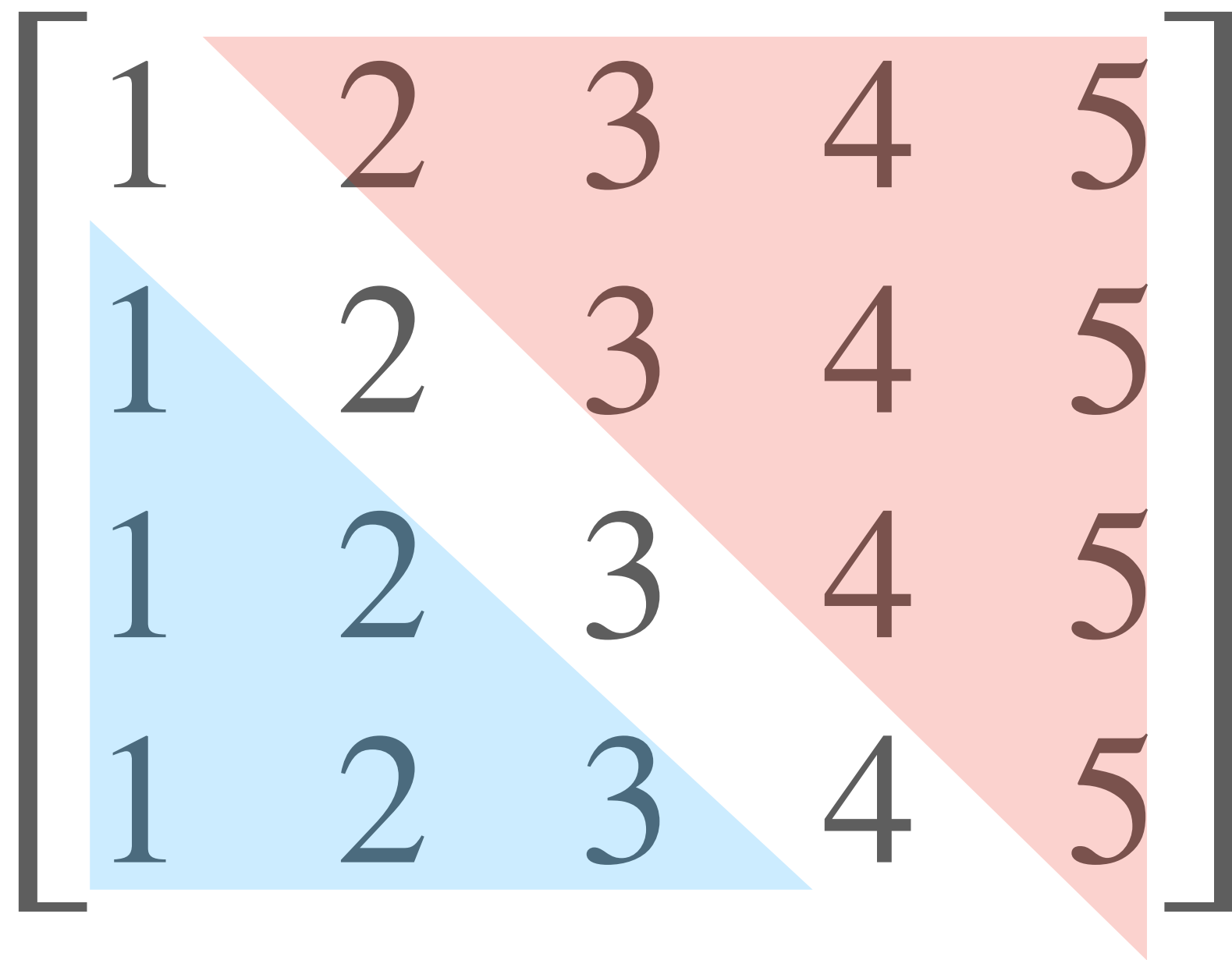
Transpose (Pictorially)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 \end{bmatrix}$$

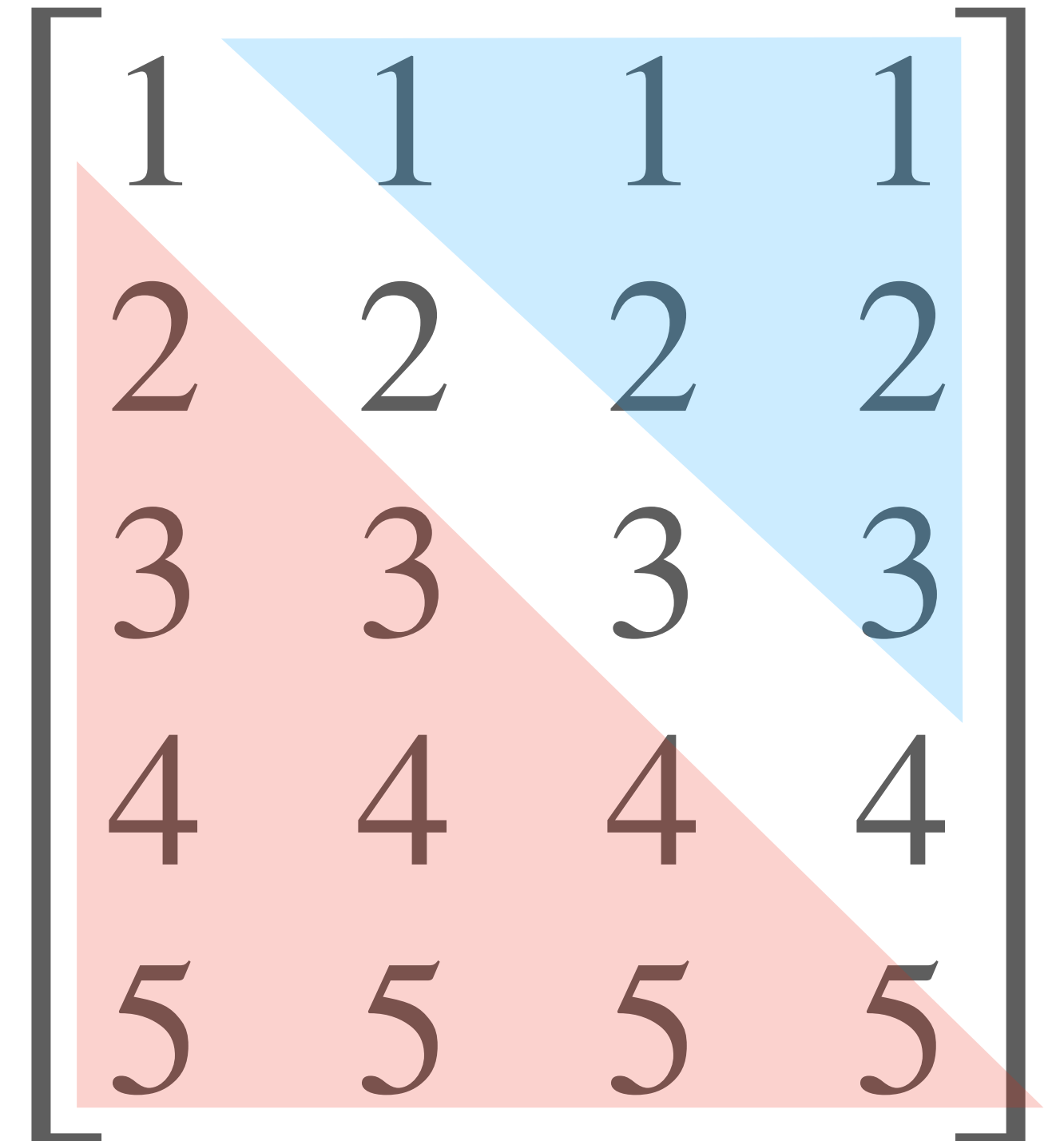
Transpose (Pictorially)



A 4x5 matrix with elements 1 through 5. The lower triangle (including the diagonal) is shaded light blue, and the upper triangle is shaded light red. The diagonal elements are 1, 2, 3, 4.

1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5

4 x 5



A 5x4 matrix with elements 1 through 5. The lower triangle (including the diagonal) is shaded light red, and the upper triangle is shaded light blue. The diagonal elements are 1, 2, 3, 4, 5.

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5

5 x 4

Transpose

Definition. For a $m \times n$ matrix A , the **transpose** of A , written A^T , is the $n \times m$ matrix such that

$$(A^T)_{ij} = A_{ji}$$

$$(\bar{A}^T)_{ij} = A_{ji}$$

Example.

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Algebraic Properties (Transpose)

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(cA)^T = cA^T \text{ (where } c \text{ is a scalar)}$$

$$(AB)^T = B^T A^T$$

Algebraic Properties (Transpose)

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(cA)^T = cA^T \text{ (where } c \text{ is a scalar)}$$

$$(AB)^T = B^T A^T \quad \text{Important: the order reverses!}$$

Challenge Problem

Prove that $(AB)^T = B^T A^T$ for $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$

$$\begin{aligned} ((AB)^T)_{ij} &= (AB)_{ji} \\ &= \sum_{k=1}^n A_{jk} B_{ki} \\ &= \dots \quad (\text{exercise}) \end{aligned}$$

Transposes and Inner Products

For $\mathbf{v} \in \mathbb{R}^n$ note that $\mathbf{v}^T \in \mathbb{R}^{1 \times n}$ is a *single-row matrix*, sometimes called a **row vector**

For two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, is $\mathbf{u}^T \mathbf{v}$ defined?

$$\underbrace{\mathbf{u}^T}_{1 \times n} \underbrace{\mathbf{v}}_{n \times 1} = \langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i * v_i$$

~~$\mathbb{R}$~~
 $\mathbb{R}$

Transposes and Inner Products

$$[u_1 \quad u_2 \quad u_3 \quad u_4] \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

Transposes and Inner Products

$$[u_1 \quad u_2 \quad u_3 \quad u_4] \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

Def. The **inner product** of two vectors **u** and **v** in $\mathbb{R}^n$ is $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$

Matrix Powers

Matrix Powers

Obs. If A is an $n \times n$ matrix, then the product AA is defined

Matrix Powers

Obs. If A is an $n \times n$ matrix, then the product AA is defined

Def. For $A \in \mathbb{R}^{n \times n}$, we write A^k for the k -fold product of A with itself

Matrix Powers

Obs. If A is an $n \times n$ matrix, then the product AA is defined

Def. For $A \in \mathbb{R}^{n \times n}$, we write A^k for the k -fold product of A with itself

What should A^0 be? (we want $A^0 A^k = A^{0+k} = A^k$)

$$A^0 = I$$

$$2^0 = 1$$

Matrix Powers (Computationally)

Use `numpy.linalg.matrix_power`

This can be *much* faster than doing a sequence of matrix multiplications (*why?*)

$$\begin{aligned} A^{16} &= (A^8)^2 = ((A^4)^2)^2 \\ &= (((A^2)^2)^2)^2 \end{aligned}$$

Final Warnings about Matrix Multiplication

Final Warnings about Matrix Multiplication

1. AB is not necessarily equal to BA , even if both are defined

Final Warnings about Matrix Multiplication

1. AB is not necessarily equal to BA , even if both are defined
2. If $AB = AC$ then it is not necessary that $B = C$

Final Warnings about Matrix Multiplication

1. AB is not necessarily equal to BA , even if both are defined
2. If $AB = AC$ then it is not necessary that $B = C$
3. If $AB = 0$ (the zero matrix) it is not necessarily the case that $A = 0$ or $B = 0$

Matrix Operations (So Far)

Matrix Operations (So Far)

transpose

$$A^T$$

Matrix Operations (So Far)

transpose

$$A^T$$

scaling

$$cA$$

Matrix Operations (So Far)

transpose

$$A^T$$

scaling

$$cA$$

addition (subtraction)

$$A + B$$

$$A + (-1)B = A - B$$

Matrix Operations (So Far)

transpose

$$A^T$$

scaling

$$cA$$

addition (subtraction)

$$A + B$$

$$A + (-1)B = A - B$$

multiplication (powers)

$$AB$$

$$A^k$$

Matrix Inverses

Recall: The Identity Matrix

Recall: The Identity Matrix

The identity matrix implements the "do nothing" transformation. For any $\mathbf{v}$,

$$I\mathbf{v} = \mathbf{v}$$

Recall: The Identity Matrix

The identity matrix implements the "do nothing" transformation. For any $\mathbf{v}$,

$$I\mathbf{v} = \mathbf{v}$$

It is the "1" of matrices. For any A

$$IA = AI = A$$

Recall: The Identity Matrix

The identity matrix implements the "do nothing" transformation. For any $\mathbf{v}$,

$$I\mathbf{v} = \mathbf{v}$$

It is the "1" of matrices. For any A

$$IA = AI = A$$

These may be different sizes

Recall: The Identity Matrix

$$\begin{array}{ccccc} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} & = & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \\ 2 \times 2 & 2 \times 4 & & 2 \times 4 & 4 \times 4 & & 2 \times 4 \end{array}$$

Recall: The Identity Matrix

Recall: The Identity Matrix

Definition. The $n \times n$ **identity matrix** is the matrix whose *diagonal* contains all 1s, and all other entries are 0s

$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$

Recall: The Identity Matrix

Definition. The $n \times n$ **identity matrix** is the matrix whose *diagonal* contains all 1s, and all other entries are 0s

$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$

Example.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Basic Algebra

$$2x = 10$$

Basic Algebra

$$2x = 10$$

How do we solve this equation?

Basic Algebra

$$2x = 10$$

How do we solve this equation?

Divide on both sides by 2 to get $x = 5$

Basic Algebra

$$2x = 10$$

How do we solve this equation?

Divide on both sides by 2 to get $x = 5$

Multiply each side by $\frac{1}{2}$ a.k.a. 2^{-1}

Basic Algebra

$$2x = 10$$

How do we solve this equation?

Divide on both sides by 2 to get $x = 5$

Multiply each side by $\frac{1}{2}$ a.k.a. 2^{-1}

$\frac{1}{2}$ is the **reciprocal** or **multiplicative inverse** of 2

Basic Algebra

$$2^{-1}(2x) = 2^{-1}(10)$$

How do we solve this equation?

Divide on both sides by 2 to get $x = 5$

Multiply each side by $\frac{1}{2}$ a.k.a. 2^{-1}

$\frac{1}{2}$ is the **reciprocal** or **multiplicative inverse** of 2

Basic Algebra

$$1x = 5$$

How do we solve this equation?

Divide on both sides by 2 to get $x = 5$

Multiply each side by $\frac{1}{2}$ a.k.a. 2^{-1}

$\frac{1}{2}$ is the **reciprocal** or **multiplicative inverse** of 2

Basic Algebra

$$x = 5$$

How do we solve this equation?

Divide on both sides by 2 to get $x = 5$

Multiply each side by $\frac{1}{2}$ a.k.a. 2^{-1}

$\frac{1}{2}$ is the **reciprocal** or **multiplicative inverse** of 2

Wouldn't it be nice...

$$\mathbf{Ax} = \mathbf{b}$$

Wouldn't it be nice...

$$\mathbf{Ax} = \mathbf{b}$$

How do we solve this equation?

Wouldn't it be nice...

$$\mathbf{Ax} = \mathbf{b}$$

How do we solve this equation?

Multiply each side by A^{-1} to get $\mathbf{x} = A^{-1}\mathbf{b}$

Wouldn't it be nice...

$$\mathbf{Ax} = \mathbf{b}$$

How do we solve this equation?

Multiply each side by A^{-1} to get $\mathbf{x} = A^{-1}\mathbf{b}$

A^{-1} is the **multiplicative inverse** of A

Wouldn't it be nice...

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$$

How do we solve this equation?

Multiply each side by A^{-1} to get $\mathbf{x} = A^{-1}\mathbf{b}$

A^{-1} is the **multiplicative inverse** of A

Wouldn't it be nice...

$$\mathbf{Ax} = \mathbf{b}$$

How do we solve this equation?

Multiply each side by A^{-1} to get $\mathbf{x} = A^{-1}\mathbf{b}$

A^{-1} is the **multiplicative inverse** of A

Wouldn't it be nice...

$$\mathbf{x} = A^{-1}\mathbf{b}$$

How do we solve this equation?

Multiply each side by A^{-1} to get $\mathbf{x} = A^{-1}\mathbf{b}$

A^{-1} is the **multiplicative inverse** of A

Do all matrices have
inverses?

Do all matrices have
inverses?

No. If they did, then every linear
system would have a solution

When does a matrix have
an inverse?

Square Matrices

Def. A matrix $A \in \mathbb{R}^{m \times n}$ is **square** if $m = n$

$$\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

i.e., it has same number of rows as columns

Square Matrices are Special

Square Matrices are Special

They are the only kind of matrices...

Square Matrices are Special

They are the only kind of matrices...

» that can have a pivot in every row and every column

Square Matrices are Special

They are the only kind of matrices...

» that can have a pivot in every row and every column

» whose transformations can be both 1-1 and onto

Square Matrices are Special

They are the only kind of matrices...

» that can have a pivot in every row and every column

» whose transformations can be both 1-1 and onto

» whose columns can have full span and be linearly independent

Square Matrices are Special

They are the only kind of matrices...

» that can have a pivot in every row and every column

» whose transformations can be both 1-1 and onto

» whose columns can have full span and be linearly independent

» *that can have inverses*

Matrix Inverses

Matrix Inverses

Def. For a matrix $A \in \mathbb{R}^{n \times n}$, an **inverse** of A is a matrix $B \in \mathbb{R}^{n \times n}$ such that

$$AB = I_n \text{ and } BA = I_n$$

Matrix Inverses

Def. For a matrix $A \in \mathbb{R}^{n \times n}$, an **inverse** of A is a matrix $B \in \mathbb{R}^{n \times n}$ such that

$$AB = I_n \text{ and } BA = I_n$$

A is **invertible** if it has an inverse. Otherwise it is **singular**

Matrix Inverses

Def. For a matrix $A \in \mathbb{R}^{n \times n}$, an **inverse** of A is a matrix $B \in \mathbb{R}^{n \times n}$ such that

$$AB = I_n \text{ and } BA = I_n$$

A is **invertible** if it has an inverse. Otherwise it is **singular**

Example. $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

Example

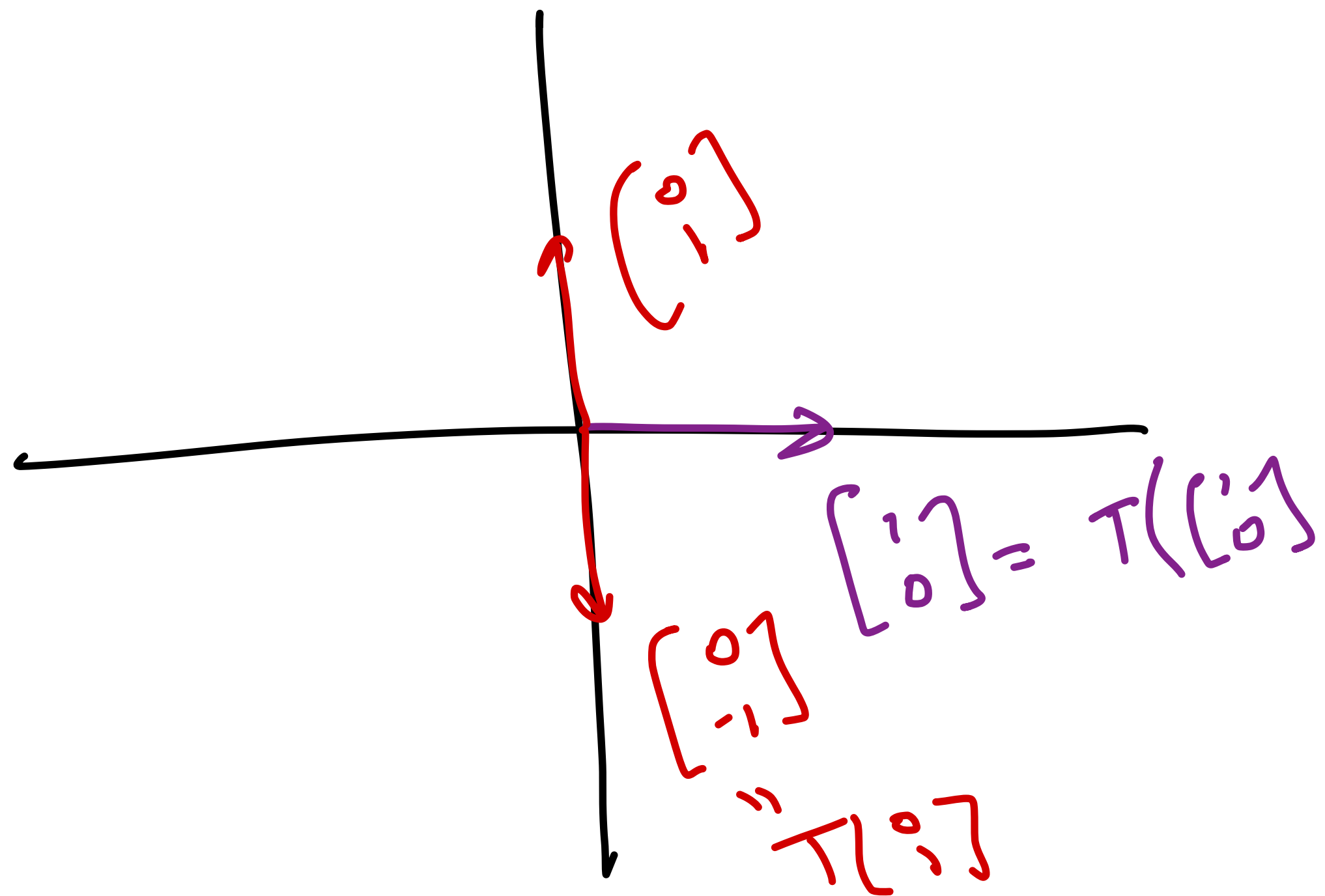
$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = (\text{exercise})$$

Example

Reflection across the x -axis is it's own inverse



$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Non-Example

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A \underbrace{B}_{3 \times 3} = I$$

$$A [\underbrace{\vec{b}_1, \vec{b}_2}_{\text{"}} \vec{b}_3] = I = [\vec{e}_1, \vec{e}_2, \vec{e}_3]$$

$$[A\vec{b}_1, A\vec{b}_2, A\vec{b}_3]$$

$$A\vec{b}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$0 = 1$$

$$\exists x. A\vec{x} = \vec{e}_3$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverses are Unique

Thm. If B and C are inverses of A , then $B = C$

Prf. Suppose B, C are inverses of A

$$B = B I = B (A C) = (B A) C = I C = C$$

If A is invertible, then we write A^{-1} for *the* inverse of A

Solutions for Invertible Matrix Equations

Thm. For matrix $A \in \mathbb{R}^{n \times n}$, if A is invertible then the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of $\mathbf{b}$

$$(A^{-1}A)\vec{v} = A^{-1}\vec{b}$$

$$I\vec{v} = A^{-1}\vec{b}$$

$$\vec{v} = A^{-1}\vec{b}$$

Unique Solutions

If $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of $\mathbf{b}$, then it has

» exactly one solution for any choice of $\mathbf{b}$

Unique Solutions

If $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of $\mathbf{b}$, then it has

» at least one solution for any choice of $\mathbf{b}$

» at most one solution for any choice of $\mathbf{b}$

Unique Solutions

If $A\mathbf{x} = \mathbf{b}$ has a unique solution for any choice of $\mathbf{b}$, then it has

» T is onto

» T is one-to-one

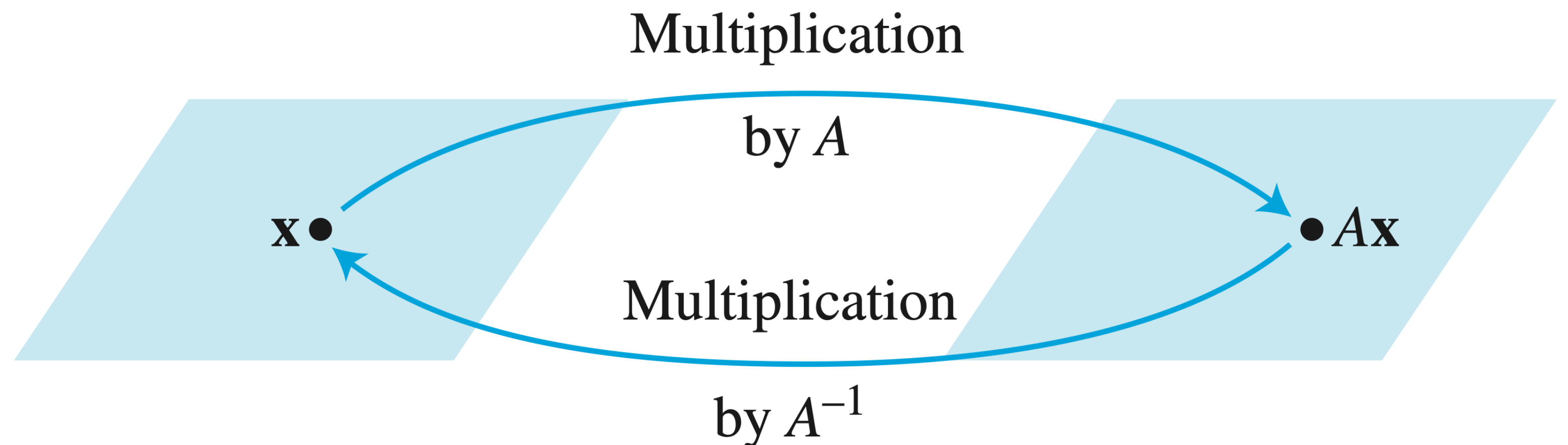
where T is implemented by A

Connection to Transformations

Def. A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is **invertible** if there is a linear transformation S such that

$$S(T(\mathbf{v})) = \mathbf{v} \quad \text{and} \quad T(S(\mathbf{v})) = \mathbf{v}$$

for any $\mathbf{v}$ in $\mathbb{R}^n$



Connection to Transformations

Connection to Transformations

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

Connection to Transformations

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

A matrix is invertible if it's possible to "undo" its transformation without "losing information"

Connection to Transformations

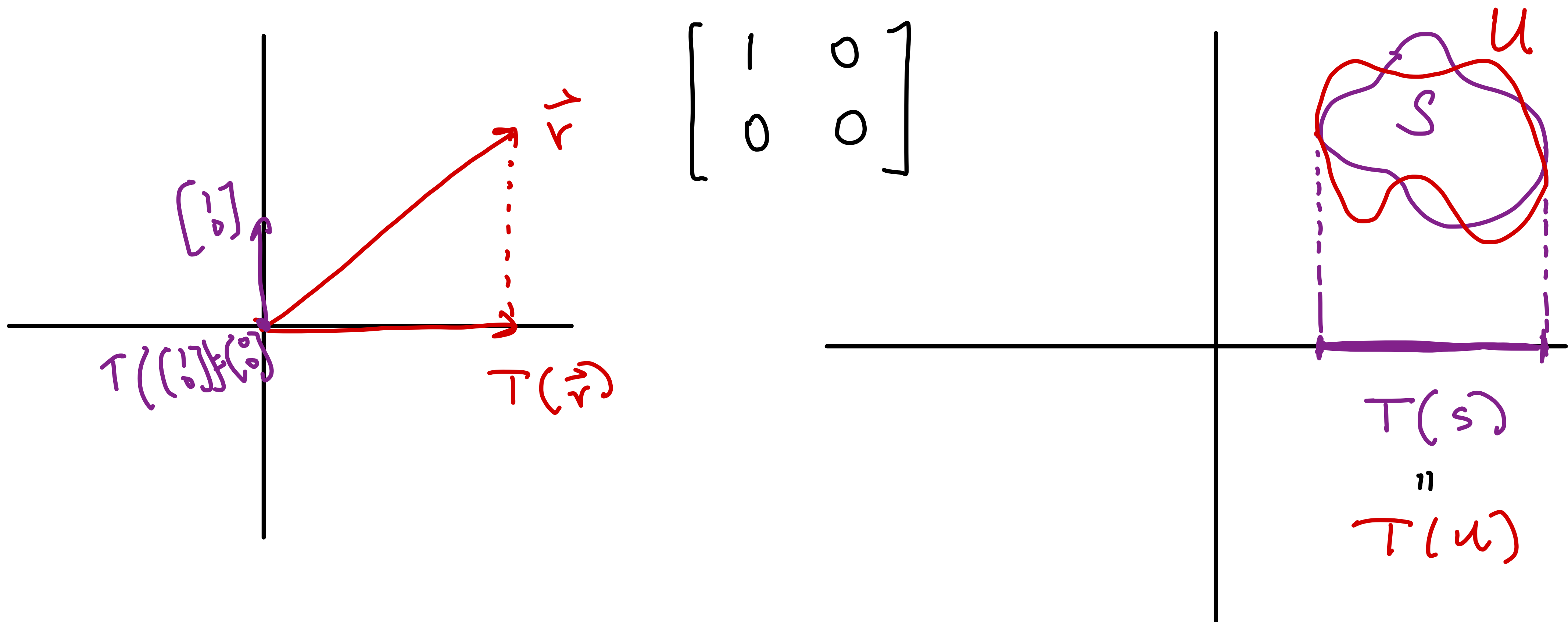
Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

A matrix is invertible if it's possible to "undo" its transformation without "losing information"

Non-Example. Projection onto the x_1 -axis in $\mathbb{R}^2$

Non-Example

Projection onto the x -axis in $\mathbb{R}^2$



Connection to Transformations

Connection to Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a **one-to-one correspondence** (bijection) if any vector $\mathbf{b}$ in $\mathbb{R}^n$ is the **image of exactly one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

Connection to Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a **one-to-one correspondence** (bijection) if any vector $\mathbf{b}$ in $\mathbb{R}^n$ is the **image of exactly one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

A transformation is a 1-1 correspondence if it is 1-1 and onto

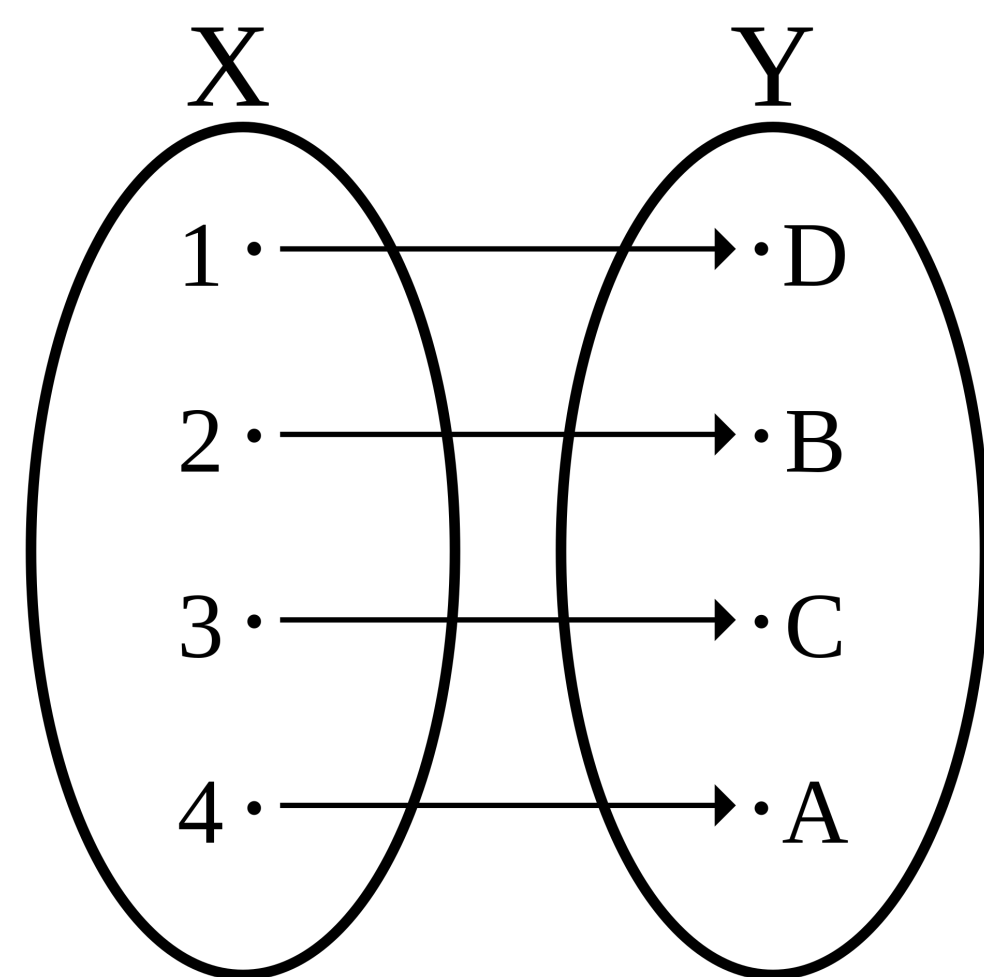
Connection to Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a **one-to-one correspondence** (bijection) if any vector $\mathbf{b}$ in $\mathbb{R}^n$ is the **image of exactly one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

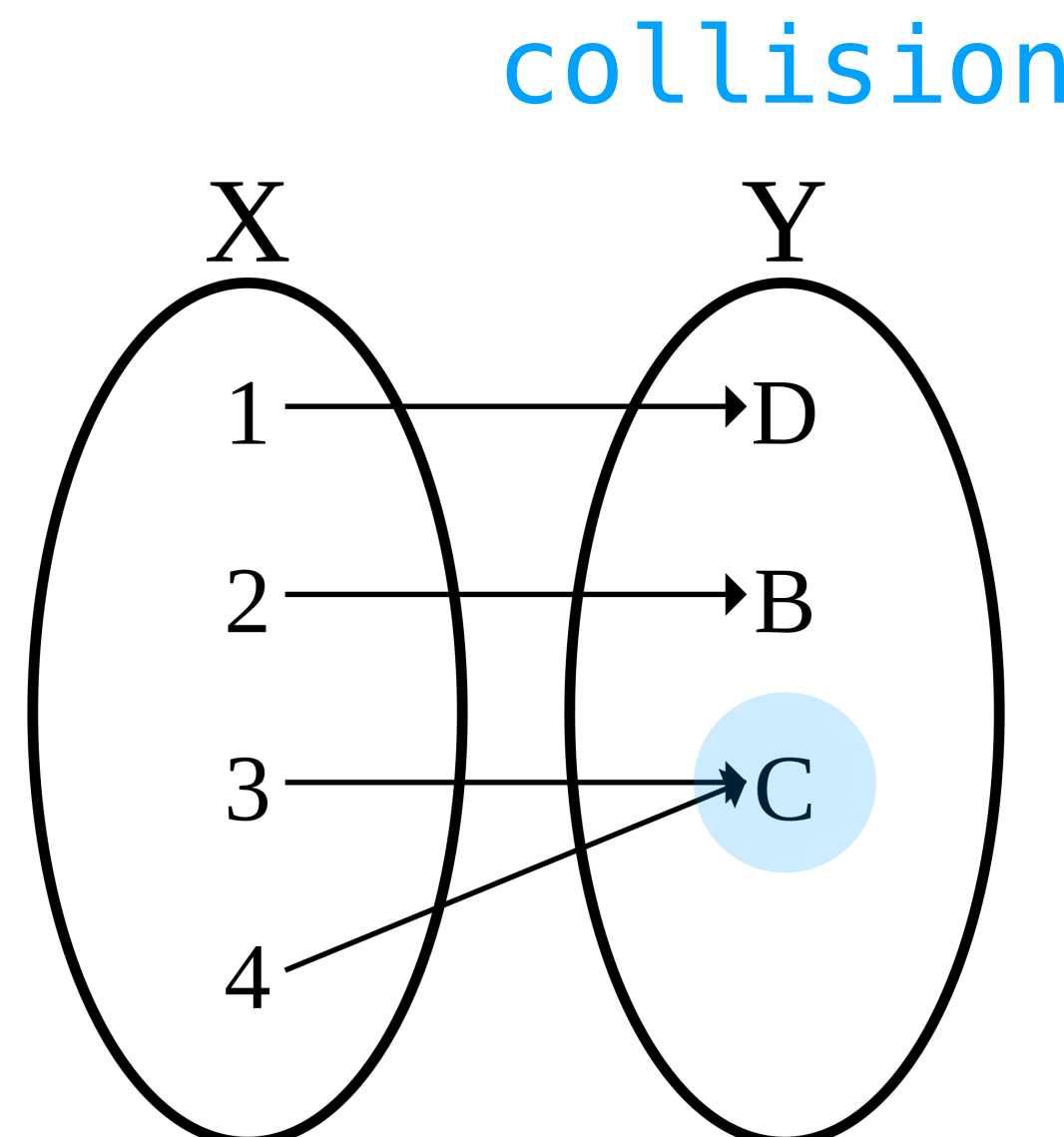
A transformation is a 1-1 correspondence if it is 1-1 and onto

Invertible transformations are 1-1 correspondences

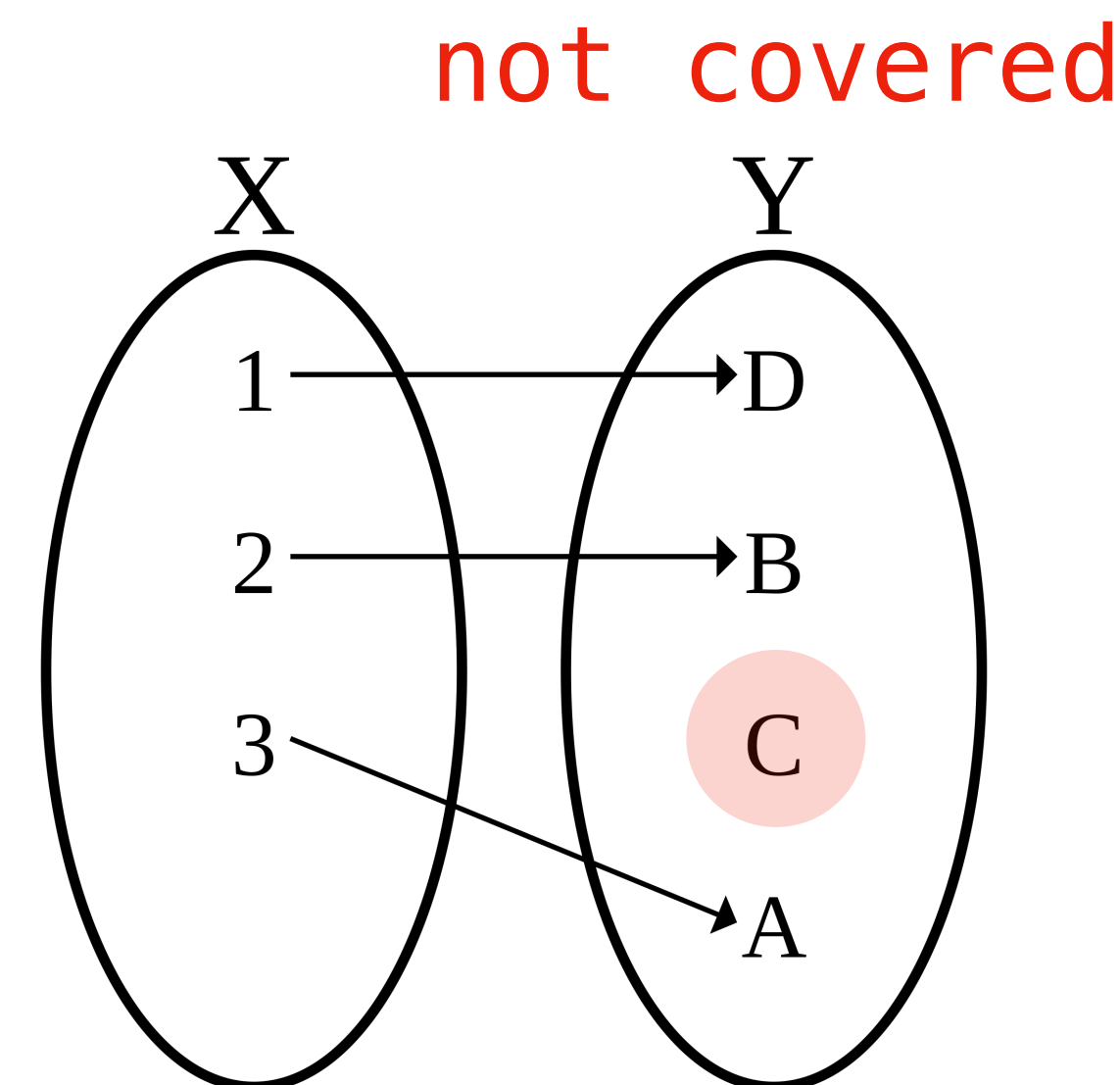
Kinds of Transformations (Pictorially)



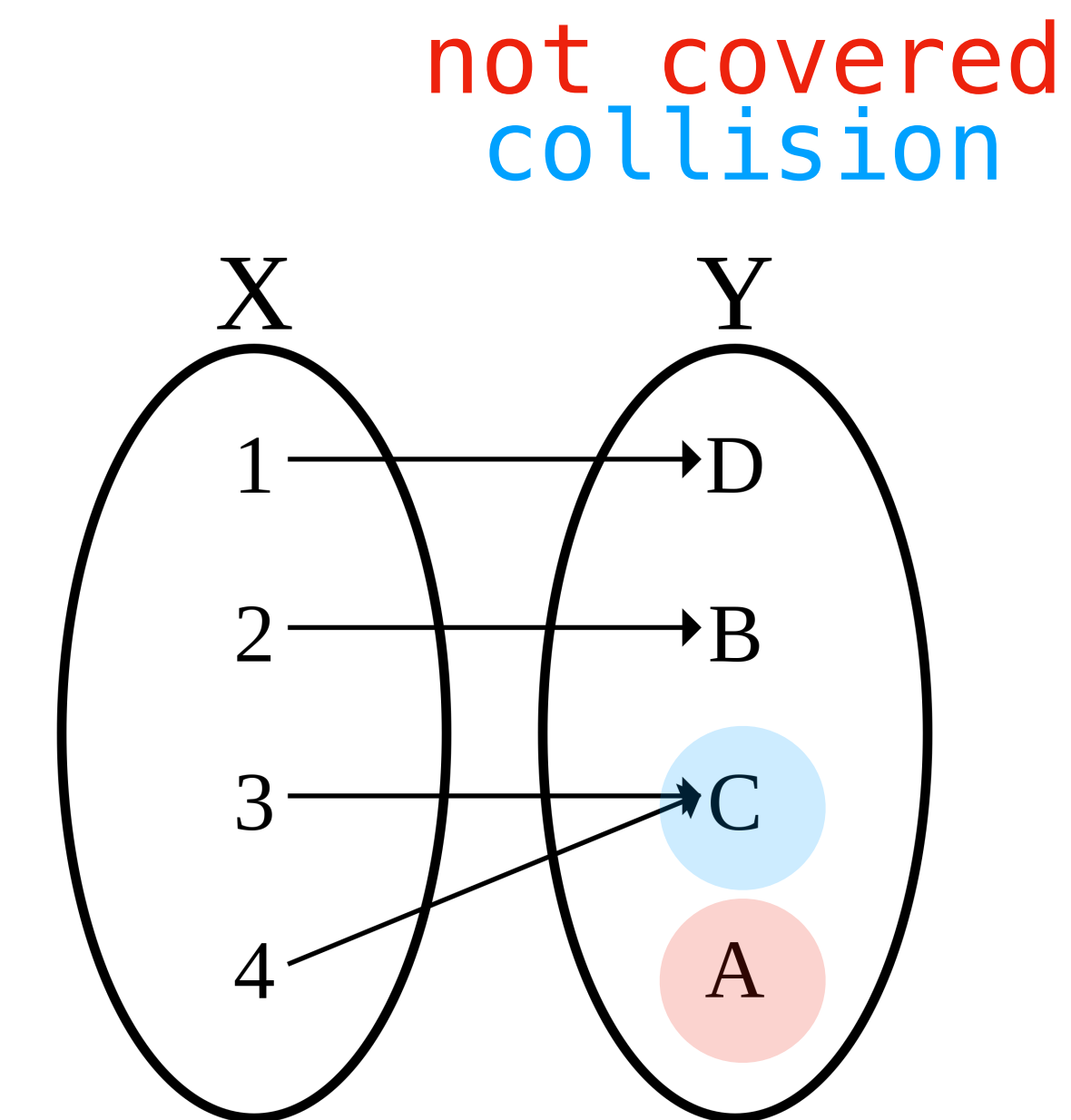
1-1 correspondence



onto, not 1-1



1-1 not onto



not 1-1, not onto

Computing Matrix Inverses

Fundamental Questions

How can we determine if a matrix has an inverse?

If a matrix has an inverse how do we compute it?

Fundamental Questions

Answer 1: Try to compute it

How can we determine if a matrix has an inverse?

If a matrix has an inverse how do we compute it?

Fundamental Questions

Answer 1: Try to compute it

How can we determine if a matrix has an inverse?

If a matrix has an inverse how do we compute it?

Answer 2: Invertible Matrix Theorem (IMT)

In General

$$A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix} = I$$

Can we solve for each $\mathbf{b}_i$?

$$A \vec{b}_1 = \vec{e}_1$$

$$A \vec{b}_2 = \vec{e}_2$$

$$A \vec{b}_3 = \vec{e}_3$$

$$\begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix}$$

In General

$$A\mathbf{b}_1 = \mathbf{e}_1$$

$$A\mathbf{b}_2 = \mathbf{e}_2$$

$$A\mathbf{b}_3 = \mathbf{e}_3$$

If we want a matrix B such that $AB = I$, then the above equation must hold (in the case B has 3 columns)

In General

$$A\mathbf{b}_1 = \mathbf{e}_1$$

$$A\mathbf{b}_2 = \mathbf{e}_2$$

$$A\mathbf{b}_3 = \mathbf{e}_3$$

If we want a matrix B such that $AB = I$, then the above equation must hold (in the case B has 3 columns)

We need to solve 3 matrix equations

How To: Matrix Inverses

Question. Find the inverse of an invertible $n \times n$ matrix A

Solution. Solve the equation $A\mathbf{x} = \mathbf{e}_i$ for every standard basis vector $\mathbf{e}_i$. Put those solutions $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_n$ into a single matrix

$$[\mathbf{s}_1 \quad \mathbf{s}_2 \quad \dots \quad \mathbf{s}_n]$$

How To: Matrix Inverses

Question. Find the inverse of an invertible $n \times n$ matrix A

Solution. Row reduce the matrix $[A \ I]$ to a matrix $[I \ B]$. Then B is the inverse of A

This is really the same thing. It's a simultaneous reduction

demo

Special Case: 2×2 Matrice Inverses

Special Case: 2×2 Matrice Inverses

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Special Case: 2×2 Matrice Inverses

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The **determinant** of a 2×2 matrix is the value $ad - bc$

Special Case: 2×2 Matrice Inverses

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The **determinant** of a 2×2 matrix is the value $ad - bc$

The inverse is defined only if the determinant is nonzero

Special Case: 2×2 Matrice Inverses

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The **determinant** of a 2×2 matrix is the value $ad - bc$

The inverse is defined only if the determinant is nonzero

(see the notes on linear transformations for more information about determinants)

Example

$$\begin{bmatrix} -6 & 14 \\ 3 & -7 \end{bmatrix}$$

$$(-6)(-7) - 14(3)$$

$$42 - 42 = 0$$

Algebra of Matrix Inverses

Algebraic Properties (Matrix Inverses)

Thm. For an invertible matrix A , the matrix A^{-1} is invertible

$$(A^{-1})^{-1} = A$$

$$A^{-1}A = I$$

Algebraic Properties (Matrix Inverses)

Theorem. For an invertible matrix A , the matrix A^T is invertible

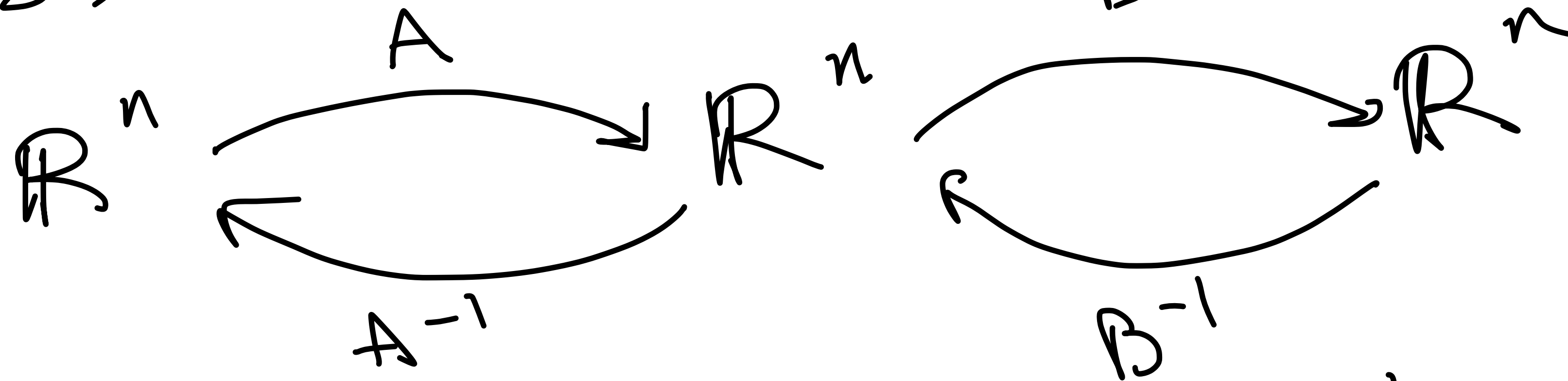
$$(A^T)^{-1} = (A^{-1})^T$$

$$A^T (A^{-1})^T = (A^{-1}A)^T = I^T = I$$

Algebraic Properties (Matrix Inverses)

Theorem. For invertible matrices $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times n}$, the matrix AB is invertible

$$(AB)^{-1} = B^{-1}A^{-1}$$



$$AB B^{-1} A^{-1} = A I A^{-1} = A A^{-1} = I$$

Question

Suppose that A is a $n \times n$ invertible matrix such that $A = A^T$ and B is a $m \times n$ matrix

Simplify the expression $A(BA^{-1})^T$ using the algebraic properties we've seen

Answer: B^T

$$A(BA^{-1})^T$$

$$A = A^T$$

Invertible Matrix Theorem

The Point

Question. How do we know if a square matrix is invertible?

Answer. *Every* perspective we've taken so far can help us answer this question

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

1. A^T is invertible

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

2. $A\mathbf{x} = \mathbf{b}$ has at least one solution for every $\mathbf{b}$
3. $A\mathbf{x} = \mathbf{b}$ has at most one solution for every $\mathbf{b}$
4. $A\mathbf{x} = \mathbf{b}$ has at exactly one solution for every $\mathbf{b}$

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

5. A has a pivot in every column

6. A has a pivot in every row

7. A is row equivalent to I_n

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

- 8. The columns of A are linearly independent
- 9. The columns of A span $\mathbb{R}^n$
- 10. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

Invertible Matrix Theorem

Thm. For any invertible matrix $A \in \mathbb{R}^{n \times n}$

- 11. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is onto
- 12. $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one
- 13. $\mathbf{x} \mapsto A\mathbf{x}$ is a one-to-one correspondence
- 14. $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

Taking Stock: IMT

The following are logically equivalent:

1. A is invertible
2. A^T is invertible
3. $A\mathbf{x} = \mathbf{b}$ has at least one solution for any $\mathbf{b}$
4. $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
5. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$
6. A has n pivots (per row and per column)
7. A is row equivalent to I
8. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
9. The columns of A are linearly independent
10. The columns of A span $\mathbb{R}^n$
11. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is onto
12. $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one
13. $\mathbf{x} \mapsto A\mathbf{x}$ is a one-to-one correspondence
14. $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

These all express the
same thing

!! only for square matrices !!

We get a lot of information for free

We get a lot of information for free

Thm. If A is square, then

A **is 1-1** if and only if A **is onto**

We get a lot of information for free

Thm. If A is square, then

A **is 1-1** if and only if A **is onto**

We only need to check one of these

We get a lot of information for free

Thm. If A is square, then

A is 1-1 if and only if A is onto

We only need to check one of these

Warning. Remember this only applies square matrices

We get a lot of information for free

Theorem. If A is square, then

A is invertible $\equiv Ax = 0$ implies $x = 0$

Invertibility is completely determined by what A sends to 0

Question

(True or False) If A is invertible, and B is row equivalent to A (we can transform B into A by a sequence of row operations), then B is also invertible

Question

If $[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$ is invertible, is $[(\mathbf{a}_1 + \mathbf{a}_2 - 2\mathbf{a}_3) \ (\mathbf{a}_2 + 5\mathbf{a}_3) \ \mathbf{a}_3]$ also invertible?

Summary

The algebra of matrices can help us simplify matrix expressions

The invertible matrix theorem connects all the perspectives we've taken so far